

Introduction to Python Programming

Welcome to the Introduction to Python Programming lecture. In this session, we will cover the fundamental concepts of Python programming, including source code, syntax, and the Python interpreter. By the end of this lecture, you will be able to write simple expressions and understand the output generated by the Python console.

What is Source Code?

Source code is the sequence of instructions that make up a program. It is written in a programming language, in this case, Python. The source code is what you, as a programmer, will write and edit to create your applications.

Understanding Syntax

Syntax refers to the set of legal structures and commands that can be used in a programming language. Each programming language has its own syntax rules that dictate how code must be written. In Python, for example, indentation is crucial as it defines the blocks of code.

The Python Interpreter

Python is an interpreted language, which means that the Python interpreter directly executes the source code without the need for a separate compilation step. This allows for an interactive environment where you can write and test code snippets quickly. The Python interpreter provides a console where you can enter commands and see immediate results.

Example of Using the Python Console

When you open the Python console, you will see a prompt (`>>>`) where you can enter expressions. For instance: - Entering `3 + 7` will output `10`. - Entering `3 < 15` will output `True`. - If you enter `'print me'`, it will return `'print me'`, while `print 'print me'` will display `print me` on the console.

Writing Simple Expressions

An expression is a combination of values and operations that computes a value. In Python, you can use various arithmetic operators: - `+` for addition - `-` for subtraction - `*` for multiplication - `/` for division - `%` for modulus (remainder) - `**` for exponentiation

Operator Precedence

Python follows a specific order of operations, known as precedence. For example, multiplication and division have higher precedence than addition and subtraction. Thus, in the expression `1 + 3 * 4`, the multiplication is performed first, resulting in `13`. You can use parentheses to enforce a specific order, such as in `(1 + 3) * 4`, which results in `16`.

Working with Numbers

Python can handle both integers and real numbers (floating-point numbers). For example: - Integer division: `35 / 5` results in `7`. - Real number division: `15.0 / 2.0` results in `7.5`. - The modulus operator can be used to find remainders, such as `14 % 4`, which results in `2`.

Using Variables

A variable is a named piece of memory that can store a value. You can assign a value to a variable using the assignment statement: - Example: `x = 5` or `gpa = 3.14`. Once a variable has been assigned a value, you can use it in expressions. For instance, if `x = 7`, then `x + 4` will yield `11`.

Outputting to the Console

To display output in Python, you can use the `print` function. The syntax is: - `print("Message")` or `print(Expression)`. You can also print multiple items on the same line by separating them with commas: - Example: `print("Hello, world!")` will output `Hello, world!`.

Getting User Input

You can read input from the user using the `input` function. For example: `- age = input("How old are you? ")` will prompt the user for their age and store it in the variable `age`.

Conclusion

In this lecture, we have introduced the basics of Python programming, including source code, syntax, and the Python interpreter. You have learned how to write simple expressions, use variables, and output results to the console. In the next lecture, we will delve deeper into control structures and data types in Python.

Quiz Questions

Q1: What is source code?

- The output of a program
- The sequence of instructions that make up a program
- The Python interpreter
- A type of variable

Correct: The sequence of instructions that make up a program

Q2: What does the Python interpreter do?

- Compiles source code into machine code
- Executes source code directly
- Translates Python code into Java
- Stores variables in memory

Correct: Executes source code directly

Q3: Which operator has higher precedence in Python?

- +
- -
- *
- %

*Correct: **